

STUDENT ACTIVITY

Workflow Sequencing — Activity

Activity: Data Preprocessing Pipeline Mapping

Q: Read the enterprise data scenario below. Map out the correct multi-step lifecycle for preparing the dataset by ordering the following five preprocessing stages from first to last:

- Detect and remove extreme values using the Interquartile Range (IQR) method and box plots
- Check for missing values and replace them with the mean or median.
- Apply OneHotEncoding to nominal categorical variables.
- Load the raw dataset using Pandas.
- Feed the processed dataset into the Machine Learning prediction model.

Next, using the provided text, contrast the specific use cases for OneHotEncoding versus LabelEncoding in three sentences. Ensure you explicitly mention the types of categorical data each is best suited for.

The Scenario: A business intelligence team is building a predictive model for customer analytics. They have received a real-world dataset containing missing entries, which occur when data is empty or unavailable. The dataset also contains nominal categorical data like item colors , ordered categorical data like customer ranking , and extreme values known as outliers that may have occurred due to incorrect measurements. Before the Machine Learning model can directly process this information to achieve high accuracy and reliability, the team must execute a structured data cleaning and encoding pipeline.

Data Preprocessing Pipeline Mapping

Correct Order of the Data Preprocessing Lifecycle

Step	Preprocessing Stage	Explanation
1	Load the raw dataset using Pandas.	Import the dataset into Python using the Pandas library so it can be analyzed and processed.
2	Check for missing values and replace them with the mean or median.	Handle missing (null) values to ensure the dataset is complete and suitable for analysis. Mean is commonly used for normally distributed data, while median is preferred when data contains outliers.

Step	Preprocessing Stage	Explanation
3	Detect and remove extreme values using the Interquartile Range (IQR) method and box plots.	Identify and remove outliers that may result from incorrect measurements or unusual observations, improving the quality of the dataset.
4	Apply OneHotEncoding to nominal categorical variables.	Convert nominal categorical variables (such as item colors) into numerical binary columns so Machine Learning algorithms can process them.
5	Feed the processed dataset into the Machine Learning prediction model.	After preprocessing is complete, use the cleaned and encoded dataset to train or test the Machine Learning model for accurate predictions.

Contrast Between OneHotEncoding and LabelEncoding

OneHotEncoding is best suited for **nominal categorical data**, where the categories have **no natural order**, such as item colors (Red, Blue, Green). It converts each category into a separate binary (0 or 1) column, preventing the Machine Learning model from assuming any relationship between the categories. **LabelEncoding** is best suited for **ordinal categorical data**, where the categories have a meaningful order, such as customer ranking (Bronze, Silver, Gold), because it assigns each category a numerical label while preserving the natural ranking.

Applying the Pipeline to the Given Scenario

The business intelligence team first **loads the raw customer dataset using Pandas**. Next, they **identify and replace missing values** with the **mean or median** to ensure data completeness. They then **detect and remove outliers** using the **Interquartile Range (IQR) method and box plots**, eliminating extreme values caused by incorrect measurements. After cleaning the data, they **apply OneHotEncoding to nominal categorical variables**, such as item colors, while **LabelEncoding is appropriate for ordered categories**, such as customer rankings. Finally, the **fully preprocessed dataset is fed into the Machine Learning prediction model**, enabling accurate, reliable, and efficient customer analytics.

Correct Preprocessing Pipeline

1. Load the raw dataset using **Pandas**.
2. Check for **missing values** and replace them using the **mean or median**.
3. Detect and remove **outliers** using the **Interquartile Range (IQR) method and box plots**.
4. Apply **OneHotEncoding to nominal categorical variables**.
5. Feed the **processed dataset** into the **Machine Learning prediction model**.

OneHotEncoding vs LabelEncoding

- **OneHotEncoding** is used for **nominal categorical data**, where there is **no natural order** (e.g., Red, Blue, Green).
- It creates **separate binary columns** for each category, preventing false numerical relationships.
- **LabelEncoding** is used for **ordinal categorical data**, where the categories have a **meaningful order** (e.g., Bronze, Silver, Gold), by assigning numerical labels while preserving the ranking.